INTRODUCTION TO

R FOR DATA SCIENCE

Comprehensive Learning Notes

Practical Guide to R Programming Fundamentals

WHAT YOU'LL LEARN

- R Programming Fundamentals
- Data Types & Variables
- Control Flow (Loops & Conditions)
- Functions & Error Handling
- Vectors, Lists, Factors & Data Frames
- File Import & Export
- Data Manipulation
- Web Scraping with R
- Regular Expressions
- RStudio & Jupyter Notebook

TOOLS & TECHNOLOGIES

{R} R Programming | >_ RStudio | [ ] Jupyter Notebook

R Console

```r
library(tidyverse)

df <- read_csv("data.csv")
summary(df)

model <- lm(y ~ x, data = df)
print(model)
```


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Module 1

1 Overview of R

1.1 Course Introduction

This course has been designed by **Yan Lou, PhD**, a data scientist and developer at IBM, Canada. He has built innovative AI and cognitive applications in areas such as:

- Mining software repositories
- Personalized health management
- Wireless networks
- Digital banking

Yan Lou earned his PhD in machine learning from the **University of Western Ontario**.

1.2 Why Choose R?

Many people wonder which programming language to learn for data science. **R is an excellent choice** for the following reasons:

1.2.1 Open-Source Language

- R is **free and open-source**, allowing users to review and modify the source code.
- Enables integration of cutting-edge AI research into analytics.

1.2.2 Rich Data Science Libraries

R offers powerful libraries for:

- **Data Processing**
- **Data Visualization**
- **Modeling**
- **Statistical Computing**

1.2.3 Simple and Easy to Learn

R has a **simple syntax**, making it easy to understand and learn quickly.

1.2.4 Strong Community Support

R benefits from a **vibrant and supportive community**, ensuring help is readily available.

1.3 Popularity of R

R has been a widely used language since **2014**. According to the **TIOBE index**:

- R ranked **8th** in **2020**.
- As of **July 2023**, R remains among the top choices for data professionals worldwide.

2 Introduction to the R Programming Language

2.1 Purpose of the R Language

R is an interpreted programming language designed for statistical computing and data analysis. It is widely used in various fields, including data science, statistical modeling, and machine learning.

2.2 Features of R

2.2.1 Interpreted Language

- R is an interpreted language, meaning it does not require compilation before execution.
- Other interpreted languages include Python and JavaScript.
- Accessed through a command-line interpreter.

2.2.2 History of R

- Developed at Bell Laboratories.
- Initially released in 1994.

2.3 Statistical Computation with R

2.3.1 Descriptive Statistics

- R is used for statistical computation, including:
    - Mean
    - Variance
    - Median
    - Quantiles
- Example: If you have a variable `numtested` representing the total number of patients tested, R can quickly return its median and maximum values.

2.3.2 Data Sampling

- Used to obtain a representative data subset.

2.3.3 Correlation Analysis

- R can compute and visualize a correlation matrix to analyze relationships between different variables.

2.3.4 Hypothesis Testing

- Used to test if the mean values of two groups are statistically different.

2.3.5 Inferential Statistics

- R helps infer properties of unknown distributions, such as estimating a population's mean from sample data.

2.4 Data Visualization in R

- R offers various visualization packages for:
    - Creating interactive and visual analytics.
    - Building dashboards for end-users.
    - Visualizing data on interactive maps, such as analyzing zip-related data.

2.5 Predictive Analysis with R

2.5.1 Machine Learning and AI

- R is a powerful tool for predictive analytics.
- Used for tasks such as:
    - Predicting pandemic trends.
    - Understanding human languages in various forms, such as documents and speeches.
    - Sentiment analysis of user reviews to determine positive or negative sentiments.
    - Visual object recognition, such as identifying dog breeds.

2.6 Summary

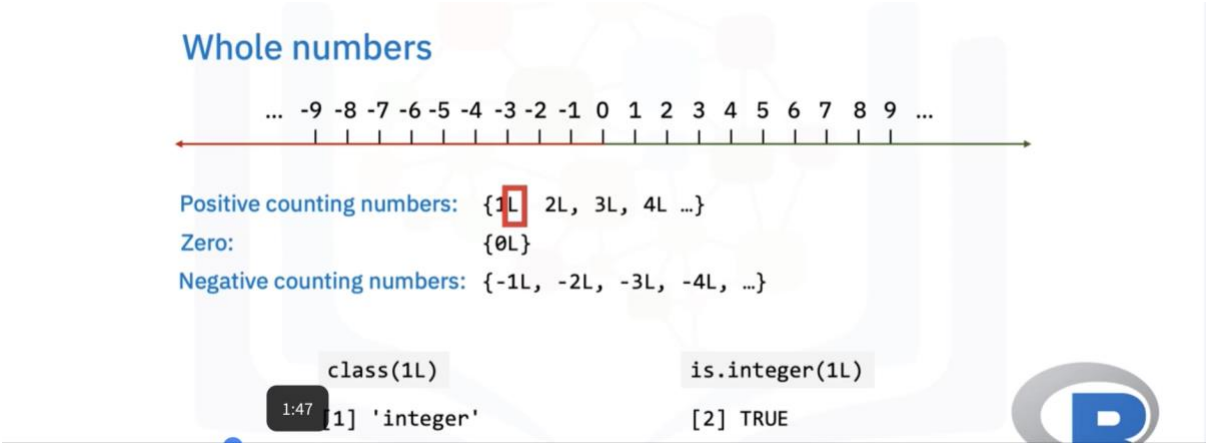
- R is an interpreted language designed for statistical computing.
- It is used for descriptive statistics, correlation analysis, hypothesis testing, and inferential statistics.
- R is widely used for data visualization and predictive analytics.
- It is a valuable tool in machine learning and AI applications.

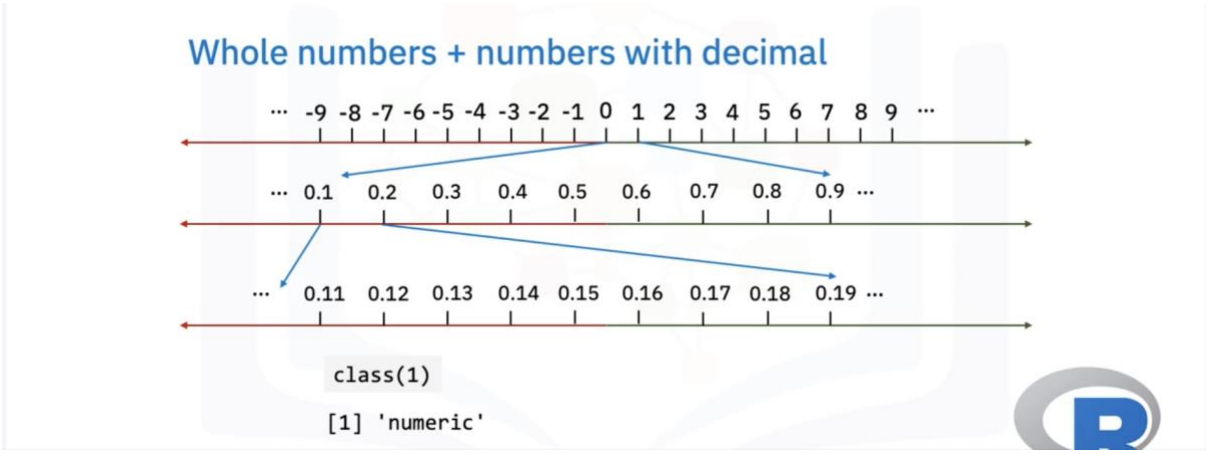
3 R Basic Data Types

- **Data Types**: R has several basic data types, including:
    - **Integer**: Whole numbers, denoted with an uppercase L (e.g., 1L).
    - **Numeric**: Includes integers and decimal numbers (e.g., 1.5).

    - **Character**: Textual data, defined with single or double quotes (e.g., "Hello").
    - **Logical**: Represents TRUE or FALSE values, used in logical operations.
- **Data Type Checking**: You can use the class() function to determine the data type and is.integer(), is.numeric(), is.character(), and is.logical() functions to check specific types.
- **Data Type Conversion**: R allows conversion between data types using functions like:
    - as.numeric()
    - as.integer()
    - as.character()
    - as.logical()
- **Advanced Data Types**: There are also complex and raw data types, though they are less commonly used.

Understanding these data types and their conversions is fundamental for processing data effectively in R.





4 Math, Variables, and Strings

- **Addition, Subtraction, Multiplication, Division, and Exponentiation**: You saw how to perform these operations using numbers, such as calculating the total time of two movies and converting minutes to hours.
- **Variable Assignment**: You learned how to store results in variables (e.g., using x for total minutes and y for hours) and how to reassign values to variables.
- **Using Meaningful Variable Names**: It's important to use descriptive names for variables to enhance code readability.
- **Order of Operations**: You were introduced to the concept of operator precedence and the use of parentheses to ensure correct calculations.
- **Creating Strings**: You learned how to create strings in R by enclosing characters in quotes.

This foundational knowledge is essential for manipulating data in R as you progress in your data science journey.

5 The R Environment

5.1 Understanding the R Environment

The R environment consists of key components such as the R Console, R script files, and workspaces. Additionally, it supports development tools like RStudio and Jupyter Notebook.

5.2 R Console

When starting with R programming, the first interaction typically happens in the **R Console**:

- It is a shell interface where users can write and execute R code interactively.
- Results of executed commands appear immediately.

5.3 R Script Files

As R programming tasks become more complex, organizing code in script files becomes necessary:

- R scripts have a **.R extension**.
- These files contain multiple lines of code that can be executed in **batch mode**.
- Regardless of where the code is written (Console or Script files), the **R interpreter** processes it into objects in memory.

5.4 Objects in R

R handles different types of objects:

- **Variables** – Store data.
- **Functions** – Code snippets designed to perform specific tasks.
- **Data** – Collections of values used in analysis.
- These objects are translated into instructions executable by the CPU.

5.5 Workspaces in R

The **workspace** in R is an environment where all created objects are stored:

- Functions, variables, and data exist in memory as part of the workspace.
- Workspaces can be saved as **RData files** and reloaded later.
- The `ls` function lists all objects in the current workspace.

5.6 Working Directory in R

R uses a **current working directory** to manage files:

- The `getwd` function retrieves the current working directory.
- If necessary, the `setwd` function changes the working directory.
- To save a workspace, use the `save.image("myworkspace.RData")` function.
- To reload a saved workspace, use the `load("myworkspace.RData")` function.

5.7 R Development Tools

To enhance R programming, two popular tools are commonly used:

5.7.1 RStudio

- The most popular **Integrated Development Environment (IDE)** for R.
- Offers features such as:
    - Syntax highlighting
    - Command history
    - Workspace management
    - Direct code execution

5.7.2 Jupyter Notebook

- A **web-based interactive code editor**.
- Useful for:
    - Writing and executing code
    - Managing data and experimental results
    - Creating educational content and prototypes

5.8 Summary

- The R Console allows interactive execution of R code.
- R script files help organize and execute code in batch mode.
- Workspaces store all objects and can be saved or reloaded using RData files.
- The working directory manages file locations.
- RStudio and Jupyter Notebook are essential tools for R development and experimentation.
- Future lessons will provide more hands-on experience with these tools.

6 Introduction to RStudio

6.1 Advantages of Using RStudio

RStudio provides several benefits for R development, making it an efficient and user-friendly Integrated Development Environment (IDE).

6.1.1 Facilitating Code Writing

RStudio enhances the coding experience through two key features:

6.1.1.1 Syntax Highlighting

- The RStudio File Editor color codes different R keywords.
- Increases readability and helps identify target variables easily.

6.1.1.2 Code Auto Completion

- RStudio suggests options while typing, including function names and documentation.
- Saves keystrokes and improves code syntax accuracy.

6.2 R Working Environment

The workspace is the core concept of the R environment, allowing easy management of objects like data, variables, and functions.

6.2.1 Visualizing Objects

- Use the `ls` function to list all objects in the workspace.
- RStudio provides a visual representation of these objects for easy access and management.
- Users can clean unnecessary objects to free up memory.

6.2.2 File Management

- R projects often contain different file types, such as scripts, data, or plots.
- RStudio includes an intuitive File Explorer for managing files.
- Users can check project structures and open files directly.
    - Example: Double-clicking addresses.csv opens the file in the File Editor.

6.3 Main User Interface Components

RStudio integrates multiple panels to streamline R development:

6.3.1 File Editor Panel

- Used for writing R code and text files.
- Features syntax highlighting and auto completion for efficient coding.

6.3.2 Console Panel

- Allows users to execute R commands and see results instantly.

6.3.3 Workspace Panel

- Displays created objects, including data, variables, and functions.

6.3.4 File Explorer Panel

- Helps manage files and other assets such as plots and packages.

6.4 Summary

- RStudio enhances R coding through syntax highlighting and auto code completion.
- The workspace provides a visual interface for managing objects and memory.
- The user interface consists of essential components like the File Editor, Console, Workspace, and File Explorer for efficient workflow management.

7 Writing and Running Code in Jupyter Notebook

7.1 Introduction to Jupyter Notebook

Jupyter Notebook is a powerful tool for writing, running, and interacting with code. It consists of a series of **cells** where users can enter different types of text, including:

- Programming code
- Markdown files
- Raw text (which can later be converted into other formats)

7.2 Executing Code in Jupyter Notebook

7.2.1 Cell Execution

- Lines of code, when executed, produce **cell output**.
- Example: Executing `print("Hello Jupyter")` displays `Hello Jupyter` in the output.

7.2.2 Execution Order and Variable Access

- Cells can be executed in a specific sequence.
- Objects and outputs from previous cells can be accessed in later cells.
- Example: Defining `x = 1` in **Cell 1** allows access to `x` in **Cell 3**.

7.3 Jupyter Notebook Kernel

- Each Jupyter Notebook requires a **kernel**, which maintains an interactive session.
- The kernel executes code and returns results.
- Jupyter supports multiple kernels, including:
  - **R** (used in this course)
  - **Python**
  - **Julia**

7.4 Benefits of Jupyter Notebook

7.4.1 All-in-One Coding Environment

Jupyter Notebook integrates essential elements needed for coding tasks and experiments, including:

- **Narration** – Provides context or instructions.
- **Code** – Implements logical tasks.
- **Data** – Used for processing tasks.
- **Visual Insights** – Includes plots, images, or videos.

7.4.2 Interactive Code and Data Exploration

- Users can explore data dynamically using different cells.
- Cells allow updates and modifications while referencing outputs from previous cells.
- Example: Loading COVID-19 data from a CSV file, displaying column names, and calculating max confirmed cases (`numconf`).

7.4.3 Exporting Jupyter Notebooks

- Jupyter Notebooks can be easily converted into formats like:
  - **PDF**
  - **HTML**

    - **LaTeX**
- This makes it convenient for presentations and sharing experiment results.

7.5 Summary

- Jupyter Notebook is made up of cells containing code, markdown, or raw text.
- It integrates narration, code, data, and visualizations in one environment.
- The kernel maintains an interactive session to execute code.
- Jupyter Notebook enhances interactivity, making it useful for teaching, learning, and data exploration.

8 Summary & Highlights

Congratulations! You have completed this lesson. At this point in the course, you know:

- You can use the R programming language to perform statistical computation, data visualization, and predictive analysis.
- The four most common data types in R are integer, numeric, character, and logical.
- You can use the class() function and the is.integer(), is.numeric(), is.character(), and is.logical() functions to determine the data type.
- You can convert some data types to other data types using the as.integer(), as.numeric(), as.character(), and as.logical() functions.
- R provides math operators that you can use to perform calculations on your data.
- Using variables in your calculations and providing them with descriptive names can help shorten your code and make it easier to read.
- You can control the order of operations using parenthesis.
- The R development environment includes the R console, R script files, and workspaces.
- Two important tools for working with R code are RStudio and Jupyter.
- RStudio features, like syntax highlighting and auto code completion, make writing code easier.
- The main components of RStudio include the File Editor, Console, Workspace, and File, Plots, and Packages Explorer.
- A Jupyter Notebook is made up of cells that can contain code, markdown files, or raw text.
- An all-in-one Jupyter Notebook contains narration, code, data, and plots, images, or videos.

Module 2

9 Vectors and Factors

- **Vectors**:
    - A vector is a one-dimensional array that can store various types of data.

    - You can create a vector using the `c()` command, for example, `c(1, 2)` or `c("Toy Story", "Akira")`.
    - Vectors can hold numerical data, character strings, and logical values (True/False).
    - You can create sequences using the colon operator, e.g., `1:10` for numbers from 1 to 10.
- **Factors**:
    - Factors are categorical variables that can take on a limited number of values.
    - They can be nominal (no order) or ordinal (with order).
    - You can convert a vector into a factor using the `factor()` function, specifying the order if needed.
- **Summary Function**:
    - The `summary()` function provides information about the structure of vectors and counts occurrences in factors.

10 Working with Vectors in R

10.1 Introduction

In this video, we will demonstrate several useful operations that can be applied to vectors in the R programming language.

10.2 Naming Vector Elements

Suppose we have a vector containing the production years of four movies. To make it easier to remember which movies these years correspond to, we can use the `names` function to map a title to each year. Once mapped, we can call our `year` vector while passing in a title in the brackets, and the output will display the corresponding year.

10.3 Finding the Number of Elements

To determine the number of elements in a vector, use the `length` function and pass the vector as input. This will return the number of elements in the vector.

10.4 Sorting a Vector

To sort a vector in ascending order, use the `sort` function while passing the vector as input. This arranges the elements from the smallest to the largest. If the `names` function is used, the titles will appear alongside the years in the output.

10.5 Finding Minimum and Maximum Values

- Use the `min` function to find the smallest number in the vector.
- Use the `max` function to find the largest number in the vector.

For example, if the years in a vector range from 1985 to 2010, min(years) will return 1985, and max(years) will return 2010.

10.6 Computing the Average of a Vector

To calculate the average value of a vector:

- Use the sum function to sum all elements and divide by the number of elements.
- Alternatively, use the mean function for a simpler approach.

Both methods will yield the same result.

10.7 Descriptive Statistics with summary

The summary function provides descriptive statistics about a vector, including:

- Minimum and maximum values
- Median
- Mean
- 1st and 3rd quartiles

10.8 Accessing Vector Elements

To retrieve an element from a vector, write the vector's name followed by the desired index in square brackets:

- vector[2] retrieves the second element.
- vector[c(2,3)] retrieves the second and third elements.
- vector[1:3] retrieves a range of elements from index 1 to 3.
- Using a negative index (e.g., vector[-1]) removes that particular element.
- Accessing an index beyond the length of the vector results in an NA (missing value).

10.9 Logical Operations on Vectors

Logical operations can be applied to each element in a vector. For example, applying a condition on a cost vector will return TRUE or FALSE for each element depending on whether the condition holds.

- To retrieve only the TRUE elements, write the condition inside square brackets: vector[vector > 10].

10.10 Handling Missing Values (NA)

Missing values (NA) frequently appear in real-world datasets for various reasons. For example, if a vector represents movie age restrictions, NA values may indicate unknown age restrictions.

10.11 Element-wise Arithmetic Operations

All arithmetic operations in R are performed element-wise:

- Multiplying two vectors results in element-wise multiplication.
- Multiplying a vector by a number scales each element accordingly.

10.12 Conversion of Vector into factor

In **R Language**, converting a **vector** into a **factor** is important because **factors** help handle categorical data efficiently. Here's why it's needed:

10.12.11. Efficient Storage & Representation

- Factors store **unique categorical values** as **levels** rather than repeating values, reducing memory usage.
- Example: Instead of storing "Male", "Female", "Male", "Female", it stores them as {1: "Male", 2: "Female"} with integer mapping.

10.12.22. Better Handling of Categorical Data

- Factors are useful in statistical modeling and data analysis where categorical variables must be treated differently from numeric data.
- Example: In regression models, R automatically treats **factors** as categorical variables instead of numerical ones.

10.12.33. Useful for Data Visualization & Analysis

- Many **R functions** (like ggplot2 and table()) expect categorical variables as **factors** for correct grouping and ordering.

10.12.44. Enables Ordered Categories (Ordinal Data)

- Factors allow for **ordered categories** (e.g., "Low" < "Medium" < "High").

10.13 Conclusion

By now, you should have a good understanding of how to access elements of a vector and apply useful operations in R.

11 Lists

- **Lists in R**: A list is a collection of objects that can contain different data types (e.g., strings, numbers, vectors).

- **Creating a List**: Use the list() function to create a list by specifying the elements.
- **Accessing Elements**: Access elements using their index in square brackets (e.g., list[2] for the second element) or by name using the dollar sign (e.g., list$name).
- **Modifying Lists**: You can add new elements, modify existing ones, or remove elements by assigning NULL to them.

12 Arrays & Matrices

12.1 Arrays:

- Structures that hold data of the same type (e.g., strings, integers).
- Can be multi-dimensional (multiple rows and columns).
- Created using the c command to form a vector, followed by the array function to define dimensions.
- Access elements using their index (row and column).

12.2 Matrices:

- Similar to arrays but strictly two-dimensional.
- Created using the matrix function, specifying the vector and the number of rows and columns.
- By default, matrices are filled by columns, but this can be changed with the byrow = TRUE parameter.
- Subsets of matrices can be accessed by specifying row and column ranges.

13 Data Frame

- A **data frame** is a structure that holds correlated information, such as movie titles and their corresponding years.
- Use the data.frame function to create a data frame, with each argument representing a column as a vector.
- Access data using the **dollar sign** symbol or by specifying the column number in square brackets.
- Retrieve individual elements by specifying the row and column numbers in square brackets.
- Use the str function to get information about the data frame's structure.
- The head and tail functions display the first and last six elements of the data frame, respectively.
- To add a new column, specify the column name in square brackets and assign a vector of values.
- Use the rbind function to add a new row.
- Delete rows using negative indexing and assign the result back to the data frame.
- Remove a column by assigning a NULL value to it.

14 Summary & Highlights

Congratulations! You have completed this lesson. At this point in the course, you know:

- A vector is a string of numbers, characters, or logical data.
- Factors (also known as categorical variables) are variables that take on a limited number of different values that can be nominal or ordinal.
- You can use R to perform operations on a vector, such as sorting the items, finding the smallest or largest number, or performing arithmetic on its values.
- Lists can store different types of data, unlike vectors, which can only store data of a single type.
- An array is a single or multidimensional structure containing data of the same type (strings, characters, or integers)
- A matrix is like an array but must be two-dimensional and can be arranged by columns or rows.
- The main difference between a data frame and other data structures, like a list, is that each variable has a vector of elements of the same type.

Module 3

15 Conditional Statements & Loops

Operator	Meaning
==	Is equal to?
!=	Is not equal to?
>	Greater than?
<	Less than?
>=	Greater than or equal to?
<=	Less than or equal to?
&	And
|	Or
!	Not
%in%	Is found in?

- **Conditional Statements**:
  - Use `if` statements to evaluate conditions (e.g., checking if a movie's release year is greater than 2000).
  - An `else` block can be added for alternative actions when the `if` condition is false.
  - Logical operators (greater than, less than, equals) are used for comparisons.

- **For Loops**:
  - Iterate through elements in a vector.
  - Can be combined with `if else` statements to control output based on conditions.
- **While Loops**:
  - Execute as long as a specified condition remains true.
  - Useful for situations where the number of iterations is unknown.

16 Functions in R

- **Definition of Functions**: Functions are blocks of code that can be reused in different parts of a program.
- **Types of Functions**:
  - **Pre-defined Functions**: Built-in functions in R or provided by packages (e.g., `mean`, `sort`).
  - **User-defined Functions**: Functions created by the user (e.g., `printHelloWorld`, `add`).
- **Function Arguments**: Functions can take arguments, and the `return` statement is used to output values.
- **Conditional Logic**: Functions can include conditional statements (e.g., `if-else`).
- **Default Values**: You can set default values for function arguments.
- **Nesting Functions**: Functions can be used within other functions.
- **Variable Scope**: Variables can be defined as global or local, affecting their accessibility outside the function.

17 Strings

- **Reading Text**: A text file is read into a variable called "summary," which holds three lines of text.
- **Character Count**: The `nchar` function counts the number of characters in a string.
- **Case Conversion**: The `toupper` and `tolower` functions convert strings to upper and lower case, respectively.
- **Character Replacement**: The `chartr` function replaces specific characters in a string.
- **String Splitting**: The `strsplit` function breaks a string into a list based on a specified delimiter, and `unlist` converts it into a character vector.
- **Sorting**: The `sort` function can be used to order elements of a character vector alphabetically.
- **Concatenation**: The `paste` function combines elements of a character vector into a single string.
- **Substring Extraction**: The `substr` function retrieves specific portions of a string, while `trimws` removes leading and trailing whitespace.
- **Negative Indexing**: The `str_sub` function from the `stringr` library allows for substring extraction by counting back from the end of the string.

18 Regular Expression

- **Regular Expressions**: Used to match patterns in strings and text, particularly useful for data analysis.
- **Email Structure**: An email can be represented as a set of characters followed by an "at sign" and another set of characters.
- **Special Characters**:
    - **Period (.)**: Acts as a wildcard, matching any character.
    - **Plus Sign (+)**: Matches one or more occurrences of the preceding element.
    - **Asterisk (*)**: Matches zero or more occurrences of the preceding element.
    - **Backslash (\)**: Used to escape special characters like the period.
- **Functions in R**:
    - **grep**: Finds matches based on a regular expression.
    - **gsub**: Replaces matched strings with a specified replacement.
    - **regexpr**: Finds matching substrings in detail.
    - **regmatches**: Extracts the matched strings.

19 Date Format

- **Date Representation**: R represents dates as the number of days since January 1, 1970, based on the **Gregorian calendar**.
- **Data Frame Structure**: You explored a data frame of Oscar-winning actors, where the "date of birth" was in a non-standard format (UNIX time).
- **Date Conversion**:
    - Use the as.POSIXct function to convert UNIX time to a full timestamp.
    - Use the as.Date function to convert timestamps or character strings into Date Class objects.
- **Date Formatting**: You can specify the date format (e.g., "Year Month Day") when converting.
- **Date Operations**: You can perform operations like subtracting dates, comparing them, and using functions like weekdays, months, and quarters to extract information.
- **Creating Date Sequences**: The seq function allows you to create sequences of dates.

20 Dealing with Bugs in R Programming

20.1 Understanding Errors in R

Errors in R occur when an operation is invalid. For example, attempting to add a character and a number will result in an error message indicating that one of the arguments is non-numeric.

20.1.1 Impact of Errors

- Errors halt code execution immediately.

- In a loop, once an error is encountered, execution stops, and subsequent iterations do not run.

20.2 Debugging in R

Debugging is the process of identifying and fixing programming bugs. In large codebases, locating errors can be challenging, making debugging an essential skill for R programmers.

20.3 Handling Errors with tryCatch

If you anticipate an error, you can use a tryCatch statement to handle it gracefully and prevent script termination.

20.3.1 How tryCatch Works

- Runs the code normally if no errors occur.
- If an error occurs, tryCatch executes custom error-handling code instead of displaying the default error message.
- Custom messages or return values can be specified within tryCatch.

20.3.2 Example

- Consider an invalid addition operation inside tryCatch.
- Instead of stopping execution, tryCatch handles the error and prints a custom message.
- If no error occurs (e.g., 10 + 10), the code executes normally.

20.3.3 tryCatch with Loops

- If an error occurs inside a loop, tryCatch handles it, but the loop does not resume execution after the error.
- This results in a single error message rather than repeated messages for each iteration.

20.4 Handling Warnings in R

Warnings indicate potential issues but do not stop code execution.

20.4.1 Example

- Passing a non-numeric value (e.g., "A") to as.integer() generates a warning.
- Warnings can be caught using tryCatch, similar to error handling.
- Custom print statements can be used to handle warnings effectively.

21 Summary & Highlights

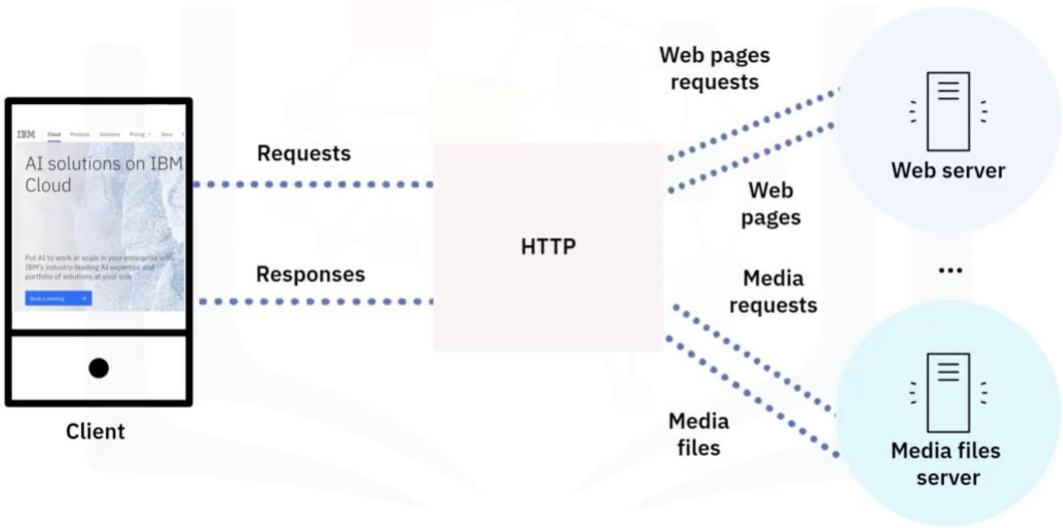
Congratulations! You have completed this lesson. At this point in the course, you know:

- If statements use comparison and logical operators to test conditions in code.
- For loops perform an operation for each item in a list, vector, or data frame column.
- While loops perform an operation until a condition is no longer true.
- Functions can be pre-defined or user-defined.
- In user-defined functions, you can control the return value of a function, add logic using if statements, and call other functions.
- You can define global variables using the <<- variable assignment operator.
- You can use R functions to manipulate the characters in a string, split a string into a vector, and retrieve specific substrings from within a string.
- Regular expressions are used to match patterns in strings and text.
- You can convert dates from one format to another using the as.POSIXct() and as.Date() functions.
- You can perform operations on Data objects using functions, like Sys.Date(), Sys.Time(), date(), and as.Date().
- You can intercept errors in R code and provide custom error and warning handling using tryCatch() statements.

Module 4

22 HTTP Request & REST APIs

- **HTTP Protocol**: The primary communication method for fetching web resources from servers. It involves a workflow where a browser sends requests to servers, which respond with the necessary resources.
- **HTTP Requests and Responses**:
    - **HTTP Request**: Comprises methods (like GET, POST, PUT, DELETE), a URL, headers, and optionally a body.
    - **HTTP Response**: Includes a status code (e.g., 200 for success, 404 for not found), headers, and a body containing the resource content.
- **REST API**: A web service that follows REST architecture, which is resource-centric, stateless, and typically uses JSON or XML for data exchange. It simplifies the design of web applications.
- **Using httr Package in R**: This package allows you to perform HTTP requests easily. You can use functions like GET() for fetching data and POST() for sending data to a server.



23 Summary & Highlights

Congratulations! You have completed this lesson. At this point in the course, you know:

- You can read a CSV file or an Excel file into a dataset using the read_csv() and read_excel() functions and then access its rows, columns, and individual data points.
- The readLines() function reads a text file into a character vector.
- The length() and nchar() functions return information about a character vector.
- The scan() function reads a text file into a character vector with each individual word as an element.
- The write()function exports a dataset as a text file.
- The write.csv() and write.table() functions export a dataset as a .csv file.
- The write.xlsx() function exports a dataset as an Excel file.
- The save() function saves R objects in .RData files.
- HTTP is a communication protocol for fetching web resources for clients from servers on the Internet and that each instance is comprised of a request and response.
- The REST API is a web service that uses the REST architecture to handle a request on a frontend web service.
- The httr package in R has functions that perform common HTTP and REST operations.
- The rvest package in R has functions that you can use to perform common web scraping tasks, such as reading HTML from a character variable, reading HTML from a URL, downloading a web page and reading it offline, extracting node data from a web page, and converting a table from a web page to a data frame.